

1 Terrain-connectivity reconstruction and its uncertainty

SWOT node water surface + Kherson gauge (EVRF2019) over the seamless terrain-bed model;
connectivity to the pre-event water network; the same-rule pre-event baseline
→ daily new and total water area, depth and volume; 1000 coherent Monte-Carlo worlds; SWOT support classes

↓ checked by

2 Independent observations

withheld gauges Kalynivske (Inhulets) and Mykolaiv (liman); Sentinel-1 per acquisition date (POD / FAR / CSI);
night ICESat-2 ground heights (pass hold-out of the class bias); SWOT vs the Kherson gauge (input consistency)

↓ disagreement explained by

3 Surface-context diagnostics

RF20 surface classes, WorldCover, ground elevation above the reconstructed surface
→ where each source is blind; S1-only detections that the connected water surface cannot reach

↓ labels, strata and blind spots inform

4 Weak-label ML diagnostics

U-Net arms on the canonical labels v004, three training seeds (v002 / v003_A as provenance)
→ behaviour under weak supervision: agreement with weak labels, not flood-mapping accuracy

Evidence runs from top to bottom: a lower level explains or diagnoses, it never overrides a higher one.
Areas carry their semantics: observed_S1 / mapped_UNet / terrain_reconstructed / literature_reported.